

INTERNAL

Planned vs deployed size, fill rate, plane progress, replenishment velocity.

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0.1s

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space/constellation_manifest

KEY FINDINGS

- Aggregate constellation fill rate: 6,429 deployed of 16,080 planned (40.0%).
- Highest fill rate: Iridium NEXT at 100.0% (75 / 75).
- Top operator by planned capacity: SpaceX (74.6% share).

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Data Sources

- `space/constellation_manifest`

Key Findings

- Aggregate constellation fill rate: 6,429 deployed of 16,080 planned (40.0%).
- Highest fill rate: Iridium NEXT at 100.0% (75 / 75).
- Top operator by planned capacity: SpaceX (74.6% share).
- Planned-capacity HHI concentration index: 5991.
- Fastest replenishment: Starlink at 2479 satellites per year.

Executive Summary

NOTE

Constellation deployment tracker measuring planned versus deployed satellite counts, orbital-plane fill progress, and replenishment velocity.

Results & Interpretation

NOTE

Planned vs deployed size, fill rate, plane progress, replenishment velocity.

Main takeaways:

- Aggregate constellation fill rate: 6,429 deployed of 16,080 planned (40.0%).
- Highest fill rate: Iridium NEXT at 100.0% (75 / 75).
- Top operator by planned capacity: SpaceX (74.6% share).
- Planned-capacity HHI concentration index: 5991.
- Fastest replenishment: Starlink at 2479 satellites per year.

Objectives

- Quantify constellation fill rates and plane progress.
- Compare operator concentration in planned capacity.
- Estimate replenishment velocity from last-launch timing.

Methodology

NOTE

Load the constellation manifest, normalize operator names, derive orbital classification, compute fill rate, plane fill, and satellites-per-year velocity, then tabulate operator share and HHI concentration.

Charts

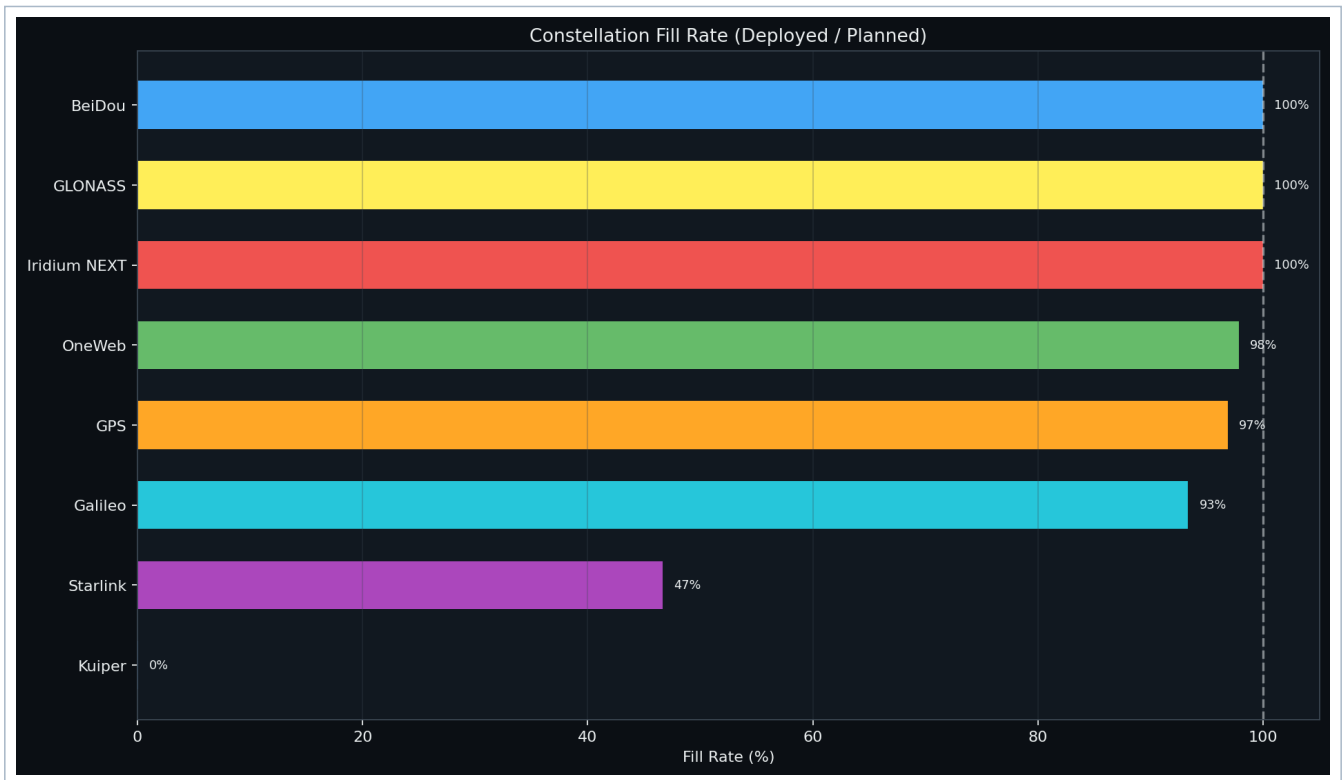


Figure 1. Constellation fill rate

Constellation fill rate

Statistical Results

constella...	operator	planned_...	deplo...	pla...	sats_per_p...	fill_rate...	plane_fill...	replenishment_v...	orbit_c...
Starlink	SpaceX	12,000	5,600	72	158	+ 46.67%	+ 49.23%	2,479.27	LEO
OneWeb	OneWeb	648	634	12	49	+ 97.84%	+ 107.82%	275.35	LEO-upper
Kuiper	Amazon/Kuiper	3,236	2	98	33	+ 0.06%	+ 0.06%	0.70	LEO
Iridium NEXT	Iridium Communications	75	75	6	11	+ 100.00%	+ 113.64%	9.88	LEO-upper
GPS	United States Space Force	32	31	6	6	+ 96.88%	+ 86.11%	2.95	MEO
Galileo	ESA	30	28	3	10	+ 93.33%	+ 93.33%	5.99	MEO
GLONASS	Roscosmos	24	24	3	8	+ 100.00%	+ 100.00%	1.62	MEO
BeiDou	China Aerospace Science And Technology Corporation	35	35	3	12	+ 100.00%	+ 97.22%	5.67	MEO

Table 1. Constellation scale metrics

operator	count_or_wei...	share_pct
SpaceX	12,000	+ 74.63%
Amazon/Kuiper	3,236	+ 20.12%
OneWeb	648	+ 4.03%
Iridium Communications	75	+ 0.47%
China Aerospace Science And Technology Corporation	35	+ 0.22%
United States Space Force	32	+ 0.20%
ESA	30	+ 0.19%
Roscosmos	24	+ 0.15%

Table 2. Operator share of planned capacity

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

constellation_scale_tracker

constella...	operator	planned_...	deplo...	pla...	sats_per_p...	fill_rate...	plane_fill...	replenishment_v...	orbit_c...
Starlink	SpaceX	12,000	5,600	72	158	+ 46.67%	+ 49.23%	2,479.27	LEO
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operator_share

operator	count_or_wei...	share_pct
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ESA	30	+ 0.19%
Roscosmos	24	+ 0.15%

Statistics Summary

Key	Value
study	constellation_scale_tracker
category	alternative_data
status	success
parameters	{ }
datasets	['space/constellation_manifest']
rows	0
git_commit	e97f0b72579223cba60ddd3e4702b560febfe339
executed_at	2026-08-04T12:27:49.875871+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-04 12:27 UTC
Author	Cerberus Research
Study slug	constellation_scale_tracker
Run ID	20260804_132749
Git commit	e97f0b72
Datasets	space/constellation_manifest

Field	Value
Data fingerprint	06dbd0623914
Duration	0.1s